

PHYS 143 – Mechanics and Thermal Physics

Lecture 9: Temperature and Heat – Supplementary Notes

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These notes support Lecture 9. They contain background that was moved out of the contact time.

1 The constant-volume gas thermometer and absolute zero

In a constant-volume gas thermometer, a fixed volume of gas is in contact with the body; its pressure p is the thermometric property. Measured at the ice point and at the steam point, the pressure is found to be a linear function of temperature. Extrapolating the straight line to zero pressure gives the same temperature, -273.15°C , whatever gas is used (at low density). This common intercept defines *absolute zero* and the kelvin scale, $T_K = T_C + 273.15$. In the kelvin scale, the ideal-gas relation reads $p \propto T$ at constant volume, which we meet in the kinetic theory lectures.

2 Thermal stress and Young's modulus

If a bar of length L_0 is prevented from expanding when its temperature rises by ΔT , the walls must compress it back by $\Delta L = \alpha L_0 \Delta T$. The tensile or compressive stress is $F/A = Y \Delta L/L_0 = Y \alpha \Delta T$, where Y is Young's modulus. For steel, $Y \approx 2 \times 10^{11} \text{ Pa}$, so a 40 K rise gives about 10^8 Pa , large enough to buckle a rail.

3 Thermal resistance

For steady conduction through a slab, $H = kA\Delta T/L$ can be written as $H = \Delta T/R$ with the *thermal resistance* $R = L/(kA)$ (unit K/W). For layers in series the resistances add, $R = R_1 + R_2 + \dots$, and the same current flows through each, which is the result we used for the copper and steel bars: $H = (T_H - T_C)/(R_1 + R_2)$.

4 Blackbody radiation in outline

A perfect absorber, a *blackbody* ($e = 1$), emits radiation with total power σAT^4 and a spectrum that peaks at a wavelength $\lambda_{\text{max}} \propto 1/T$ (Wien's law). A body at 300 K radiates mostly in the infrared; the Sun, at about 5800 K, radiates mostly in the visible. The T^4 dependence means that doubling the kelvin temperature increases the radiated power sixteen-fold.

5 Why the specific heat is not exactly constant

We treated c as constant over the temperature range. In reality c varies slowly with temperature (strongly near absolute zero). For the ranges in this lecture the average value is a good approximation. The molar heat capacity of simple solids is close to $3R \approx 25 \text{ J/(mol K)}$, as the copper example showed.

6 Heat and internal energy

Heat Q is energy transferred between systems because of a temperature difference. It is not a property of a single body, whereas the *internal energy* of a body is. Doing work on a system (for example, by friction or by compression) can also raise its temperature, which is the subject of the first law of thermodynamics.